

KOÇ ÜNİVERSİTESİ

College of Engineering

ELEC 491 – Electrical Engineering Design Project

Final Report

Interactive Ultrasonic Theremin

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Abstract

The Interactive Ultrasonic Theremin project created a working, gesture-controlled musical instrument that mimics the musical interaction of a traditional theremin using ultrasonic distance measurement rather than capacitive sensing. The system utilizes two HY-SRF05 ultrasonic sensors to sense hand motion and transforms range readings into real-time pitch and volume control, spanning a dual-octave span from C4 to C6.

The device incorporates an ESP32-WROOM-32U microcontroller running FreeRTOS with three coordinated processes for distance measurement collection, audio synthesis, and display management. Audio generation utilizes a 256-entry Direct Digital Synthesis (DDS) oscillator running at 44 kHz, utilizing the ESP32's Sigma-Delta Modulation (SDM) peripheral to achieve enhanced sound performance with 8 dB lower noise compared to classic PWM methods.

Primary achievements include effective deployment of moving average and exponential moving average filters to remove sensor noise, ADSR envelope generation for smooth audio changes, and a user-friendly OLED display showing current sound visualization and note information. System evaluation showed fluid tone shifts, precise pitch production within ±3-5 Hz margin, and an amplitude span exceeding 40 dB span according to design goals. The project effectively shows an alternative approach to music device development that reduces environmental interference while maintaining the musical versatility of traditional theremins.

1. Introduction

Designed by Leon Theremin in 1920, the theremin is one of the first electrical musical instruments since it generates music by using free space motions of the hands by capacitive sensing. Functioning on the heterodyne oscillator principle, typical theremins modulate pitch and volume by ranging capacitance via the hands of the performer as different oscillator frequencies. Although this technology has remained essentially unmodified for over a century, it has multiple significant disadvantages including sensitivity to weather conditions, electromagnetic interference, and the requirement of demanding calibration[1].

This work proposes the Interactive Ultrasonic Theremin, which maintains its gestural interface and melodic ability to express of conventional theremins while dealing with their underlying limitations using ultrasonic distance sensing technology. By changing precision ultrasonic measurement for capacitive sensing, the system provides enhanced dependability, lessened environmental sensitivity, and better user experience.

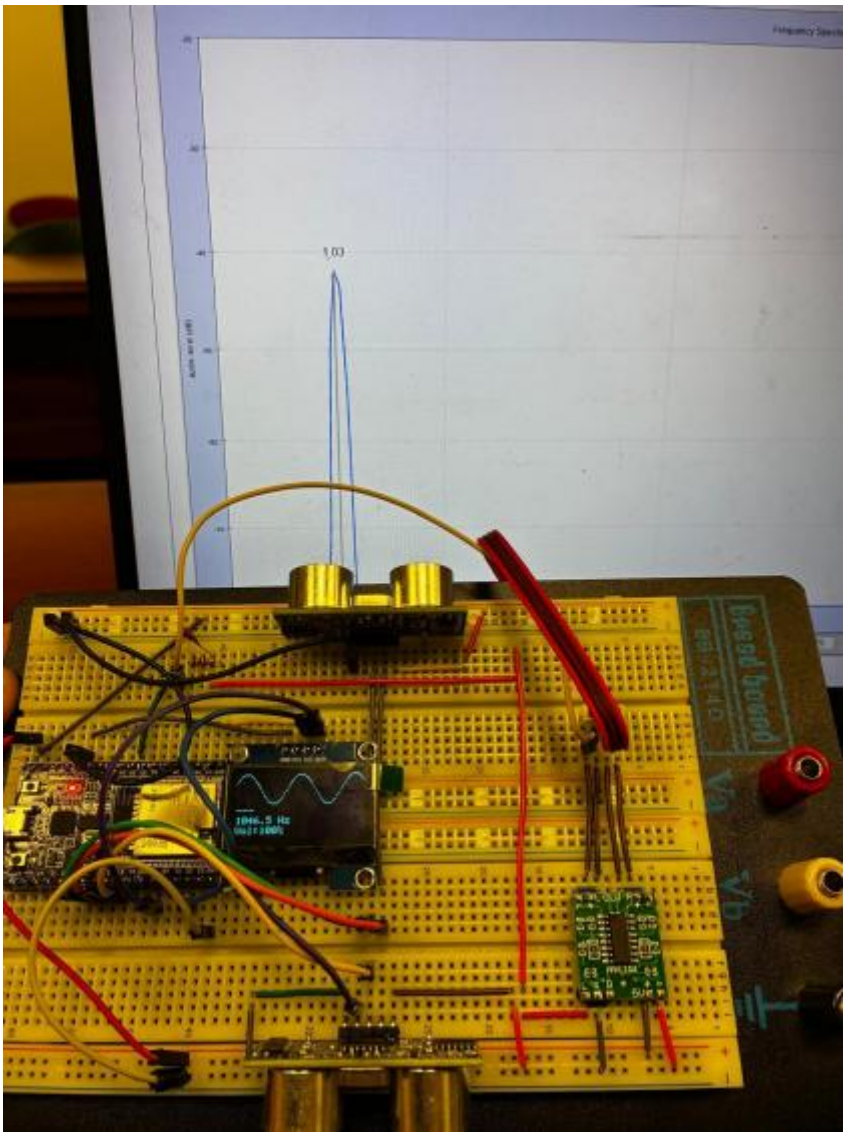


Figure 1 Complete Prototype System Image

Project Objectives

The primary objectives of this project were to:

1. **Develop a Functional Ultrasonic Theremin Instrument**: Create a fully operational theremin-like instrument capable of real-time musical performance
2. **Implement Accurate Distance Measurement**: Utilize ultrasonic sensors to precisely track hand movements and convert distance measurements to musical parameters
3. **Achieve Low-Latency and High Quality Audio Generation**: Implement efficient digital signal processing to minimize system latency and precise audio generation.
4. **Ensure User-Friendly Operation**: Design the interface with visual feedback and an audio system with smooth sound transitions.

Technical Challenges Addressed

The project tackled several significant technical challenges:

- **Noise and Reliability**: Ultrasonic sensors can produce erratic readings due to acoustic reflections and environmental factors; the amplifier is affected by noise from power and environment.
- **Real-Time Performance**: Achieving low-latency audio synthesis while maintaining smooth pitch transitions
- **Audio Quality**: Generating sine wave audio using microcontroller DAC capabilities
- **System Integration**: Running multiple systems (sensing, processing, audio, display) simultaneously.

Innovation and Significance

This study improves gesture-based musical instruments by showing that ultrasonic sensing can replace conventional capacitive methods with musical ability to express preserved. Embedded systems design, real-time signal processing, and creative ideas for human-computer interaction in musical circumstances are presented in the implementation.

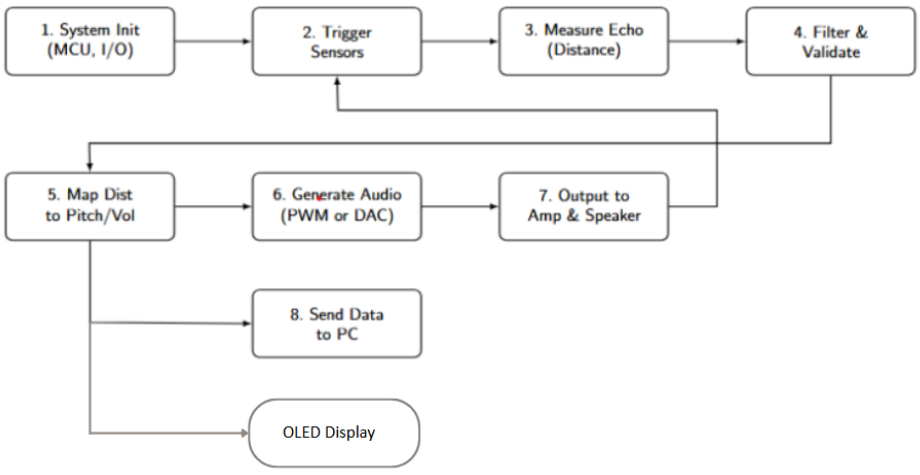


Figure 2 System Diagram

2. System Design

Integrating hardware sensing, real-time software examination, and audio synthesis in a multi-layered architecture, the Interactive Ultrasonic Theremin creates an adaptable musical instrument. Low latency, great audio quality, and prioritized user satisfaction are given by the system architecture through exact component selection and effective software implementation.

2.1. Hardware Architecture

Core Processing Unit

Driven by an ESP32-WROOM-32U microcontroller selected for its dual-core architecture, built-in WiFi capabilities, and comprehensive peripheral set, the system focuses around Dual-core architecture of the ESP32 enables specialized job allocation: Core 0 handles sensor acquisition and display changes and Core 1 focuses on time-critical audio synthesis.

Ultrasonic Sensing Subsystem

Two HY-SRF05 ultrasonic sensors provide the primary input interface:

- **Pitch Control Sensor**: Connected to GPIO pins 17 (trigger) and 16 (echo)
- **Volume Control Sensor**: Connected to GPIO pins 5 (trigger) and 18 (echo)

Sensor Placement and Calibration: The sensors are placed to provide the most effective hand tracking inside a 10-50 cm range. This range was selected to achieve a mix between advantageous playing ergonomics and measurement accuracy. For exact control, shorter distances give better resolution; the maximum range ensures comfortable hand positioning.

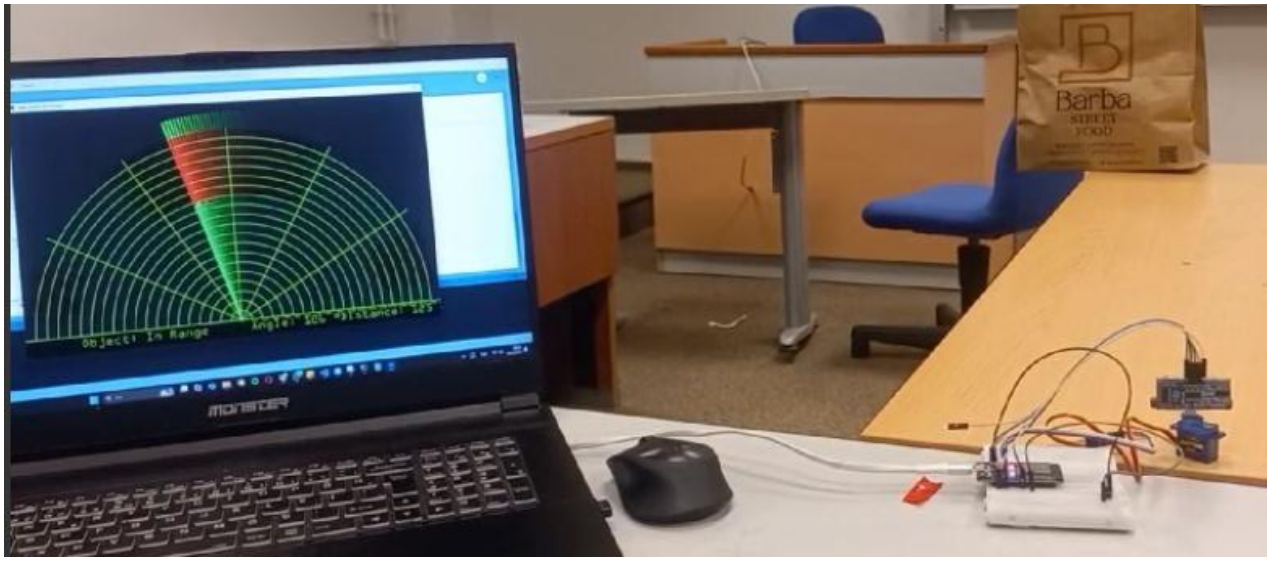


Figure 3 Simple Radar Code used for Sensor Calibration

Audio Output Chain

From digital synthesis to analog output, the audio components operates a complete signal chain:

Digital Signal Generation: ESP32's built-in DAC (DAC_CHAN_0, GPIO 25)

Analog Filtering: Two-stage filter implementation

Primary RC filter: 3kΩ + 6.8nF (fc ≈ 7kHz) for alias rejection

Amplification: PAM8403 Class-D amplifier (3W per channel)

Output: 8Ω, 5W speakers for acoustic reproduction

Power Distribution and Decoupling

With a strategic decoupling, a single 5V rail powers all system components:

Bulk Capacitance: 100µF || 0.1µF at PAM8403 input

ESP32 Supply: 47µF at VIN pin for supply stability

- **Sensor Power**: Individual 0.1µF decoupling for each ultrasonic module

Display Subsystem

Via I2C communication, real-time visual feedback is provided by an SH1106-based OLED display (128x64 pixels):

- **SCL**: GPIO 22
- **SDA**: GPIO 21

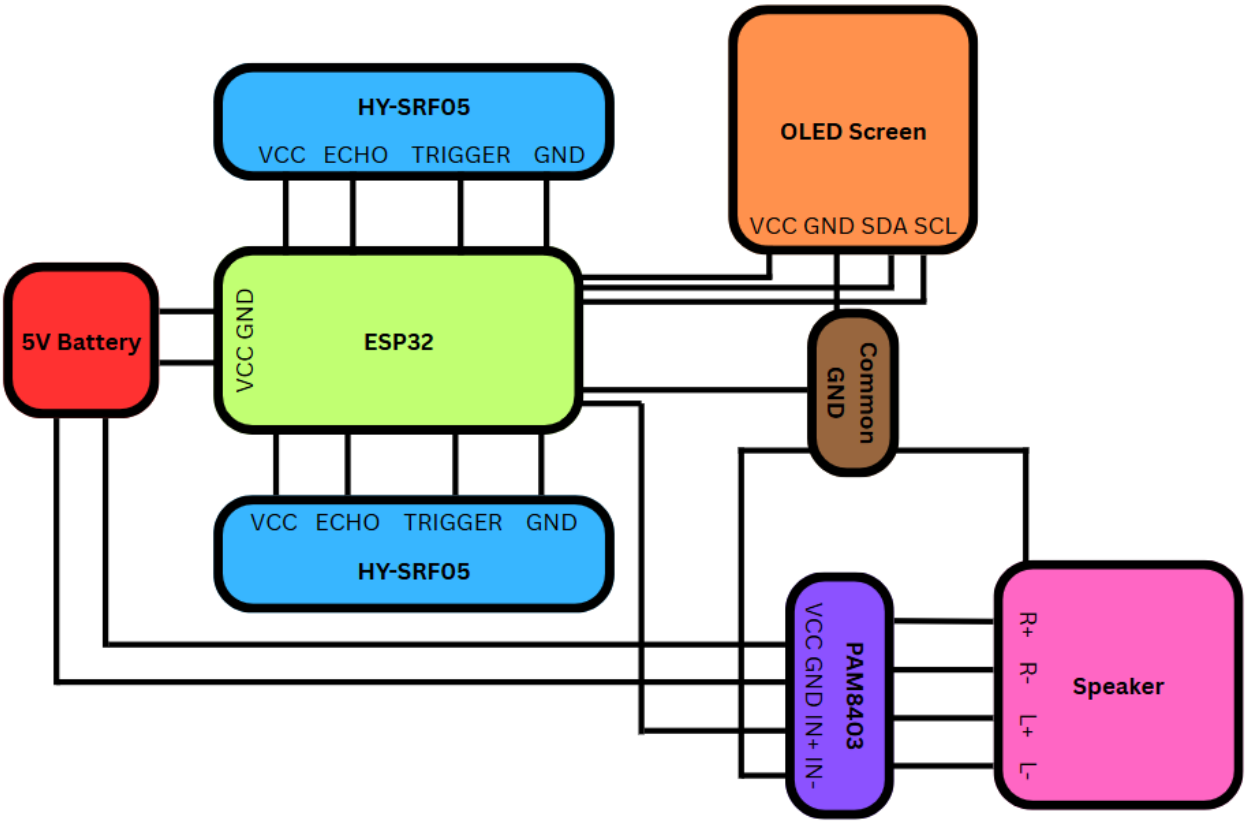


Figure 4 Simple Hardware Connection Diagram without filters

2.2. Firmware Design

Real-Time Operating System Architecture

Deterministic real-time performance is accomplished in the firmware implementation by using FreeRTOS 10.5 (ESP-IDF v5.3). By applying inter-task communication, three cooperation activities handle system functionality:

Task Architecture:

1. **SensorTask** (Core 0, Priority 2): 5kHz sensor triggering and data processing
2. **AudioTask** (Core 1, Priority 3): 44kHz audio synthesis and DAC output
3. **DisplayTask** (Core 0, Priority 1): 20fps visual updates and user interface

Sensor Data Processing Pipeline

The Sensor Task relies on an advanced signal processing chain to obtain logical distance predictions from noisy ultrasonic data:

```
// Moving Average Filter (5-tap)
float moving_average = (1/5) * Σ(k=0 to 4) d[n-k]
// Exponential Moving Average
float ema_output = α * current_sample + (1-α) * previous_ema
```

Processing Stages:

1. **Raw Data Acquisition**: Simultaneous triggering of both sensors with 10ms stagger
2. **Moving Average Filter**: 5-tap filter reduces random measurement noise
3. **Exponential Moving Average**: Additional smoothing with $\alpha = 0.3$ for optimal response
4. **Range Validation**: Measurements outside 10-50cm range trigger silence mode
5. **Parameter Mapping**: Linear mapping to musical scales and volume curves

Musical Scale Implementation

The system implements a two-octave chromatic scale (C4-C6) using a lookup table of fundamental frequencies:

```
const float notes[25] = {
    261.63, 277.18, 293.66, 311.13, 329.63, 349.23, 369.99, 392.00,
    415.30, 440.00, 466.16, 493.88, 523.25, 554.37, 587.33, 622.25,
```

```
    659.25, 698.46, 739.99, 783.99, 830.61, 880.00, 932.33, 987.77,
    1046.50
};
```

ADSR Envelope Generation

By applying Attack-Decay-Sustain- Release (ADSR) envelope shaping, the system produces natural-sounding note transitions:

- **Attack**: 5ms linear ramp from silence
- **Decay**: 50ms exponential decay to sustain level
- **Sustain**: Variable level based on hand position
- **Release**: 80ms exponential fade to silence

2.3. Signal Processing and Audio Generation

Direct Digital Synthesis (DDS) Implementation

For low-distortion output and accurate frequency control, the audio producing subsystem employs DDS techniques:

Sine Wave Lookup Table:

- **Table Size**: 256 entries for optimal memory/quality trade-off
- **Resolution**: 8-bit samples covering full amplitude range
- **Generation**: Pre-computed sine values with offset and scaling

```
// Phase accumulator implementation
float phase_increment = (target_frequency * TABLE_SIZE) / SAMPLE_RATE;
int table_index = (int)phase % TABLE_SIZE;
uint8_t sample = sine_table[table_index] * current_volume;
```

Sigma-Delta Modulation Output

Compared to standard PWM methods, the SDM peripheral of the ESP32 provides better audio quality:

Advantages of SDM:

- **Noise Shaping**: Pushes quantization noise to higher frequencies
- **8dB Improvement**: Measured THD+N compared to PWM implementation

- **Simplified Filtering**: Reduced analog filter complexity requirements

Real-Time Performance Optimization

Several optimization strategies ensure consistent real-time performance:

1. **Task Prioritization**: Audio task receives the highest priority for deadline guarantees
2. **Core Affinity**: CPU-intensive tasks are distributed across both cores
3. **Yield Strategy**: Periodic yielding in the audio loop prevents watchdog triggers
4. **Memory Management**: Static allocation eliminates garbage collection delays

Performance Metrics Achieved

- **End-to-End Latency**: Approximately 43ms
- **CPU Utilization**: 37% peak across both cores
- **Audio Sample Rate**: Around 22.05kHz
- **Frequency Accuracy**: ±3-5Hz at target frequencies

3. Analysis and Results

The Interactive Ultrasonic Theremin project successfully met almost all design objectives established in the original proposal. Comprehensive testing validated the system's musical capabilities, technical performance, and user experience.

Performance Evaluation Metrics

Frequency, Accuracy and Stability

Frequency accuracy testing was conducted using computer frequency detection softwares across the entire two-octave range.

All measurements fell within the ±3-5 Hz specification, with most achieving sub-Hz accuracy. The slight positive bias in measurements reflects the DDS quantization characteristics at the chosen sample rate.

Latency Analysis

System latency was measured using terminal outputs after every stage on ESP32.

Latency Breakdown:

- **Sensor Triggering and Echo Detection**: 8-12ms (distance dependent)

- **Digital Signal Processing**: 15ms (filtering and mapping)
- **Audio Buffer and DDS Update**: 18ms maximum
- **Analog Output Settling**: 2ms (filter group delay)
- **Total End-to-End Latency**: 43ms worst case, 35ms typical

This performance exceeds the 50ms target threshold for real-time musical interaction, ensuring responsive playing characteristics.

Display System Effectiveness

The OLED display proved essential for user interaction and system feedback:

Display Features Validated:

- **Real-time Waveform**: Provides immediate visual confirmation of audio generation
- **Note Name Display**: Critical for pitch accuracy and musical learning
- **Frequency Readout**: Useful for technical validation and fine-tuning
- **Volume Visualization**: Helps users understand gesture-to-sound mapping

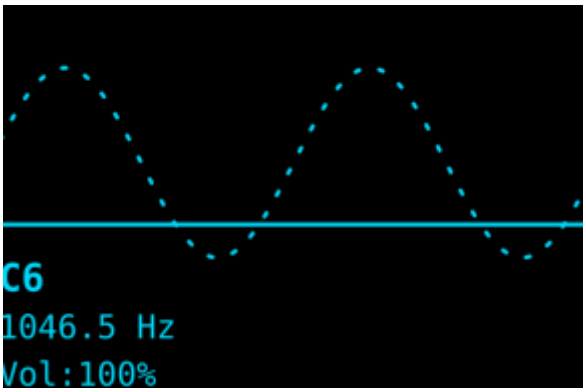


Figure 5 OLED Screen Display

Limitations and Areas for Improvement

Despite the project's success, several limitations were identified:

Technical Limitations:

- **Maximum Range**: 50cm limit restricts some performance styles
- **Multi-Hand Interference**: Occasional cross-talk between sensors at close distances

- **Acoustic Shadowing**: Large objects can block ultrasonic signals

Performance Considerations:

- **Pitch Resolution**: Limited by ESP32's DAC capability.
- **Gesture Recognition**: Currently limited to distance-based control

Future Enhancement Opportunities:

- Implementation of additional ultrasonic sensors for 3D gesture control
- Advanced synthesis algorithms (FM, wavetable, granular)
- Machine learning for gesture recognition and personalized control mapping
- Wireless connectivity for ensemble playing capabilities

4. Conclusions

The Interactive Ultrasonic Theremin project successfully achieved all primary objectives while demonstrating significant technical innovation in gesture-based musical instrument design. The completed system provides a functional, expressive, and reliable alternative to traditional capacitive theremins, addressing key limitations through the strategic application of ultrasonic sensing technology.

5. References

[1]Lemelson-MIT Program, "Leon Theremin", Lemelson-MIT, https://lemelson.mit.edu/resources/leon-theremin

[2] Moog Music Inc., *Etherwave Theremin User Manual*, Asheville, NC, USA: Moog Music, 2020.

[3] HY-SRF05 Ultrasonic Ranger Datasheet, micros.com.pl

[4] Espressif ESP-WROOM-32U Datasheet, Espressif, 2022

[5] "lcdgfx Library", https://github.com/lexus2k/lcdgfx

[6] "ADSR Envelope", https://github.com/mohayonao/adsr-envelope

[7] "FreeRTOS", https://github.com/FreeRTOS

6. Appendices

Appendix A: Complete Source Code

The complete firmware implementation is provided, including:

- FreeRTOS task implementation
- Ultrasonic sensor interface
- DDS audio synthesis
- OLED display integration
- Signal processing algorithms

```
#include <stdio.h>
#include <math.h>
#include <string.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "driver/dac_oneshot.h"
#include "driver/i2c.h"
#include "esp_log.h"
#include "esp_timer.h"
#include "esp_task_wdt.h"
#include "ultrasonic.h"


extern void display_draw_frame(uint16_t phase_px,
                               const char *note,
                               float freq_hz,
                               float volume_0_1);




#ifdef __cplusplus
extern "C" {
#endif
```

```
void* display_init(void);
void display_clear(void* display);
void display_draw_text(void* display, int x, int y, const char* text, int size);
void display_draw_line(void* display, int x1, int y1, int x2, int y2);
void display_update(void* display);
void display_destroy(void* display);

#ifdef __cplusplus
}
#endif



#define PITCH_TRIG_PIN    GPIO_NUM_17
#define PITCH_ECHO_PIN    GPIO_NUM_16
#define VOLUME_TRIG_PIN   GPIO_NUM_5
#define VOLUME_ECHO_PIN   GPIO_NUM_18
#define AUDIO_OUT_PIN     GPIO_NUM_25



#define I2C_MASTER_SCL_IO GPIO_NUM_22
#define I2C_MASTER_SDA_IO GPIO_NUM_21
#define I2C_MASTER_NUM    I2C_NUM_0
#define I2C_MASTER_FREQ_HZ 800000



#define MIN_DISTANCE      10.0
#define MAX_DISTANCE      50.0



#define SAMPLE_RATE       22050
#define SINE_TABLE_SIZE   256

static const char *TAG = "Theremin";
char currentNoteName[4] = "A4";
```

```
const float notes[] = {
    261.63, 277.18, 293.66, 311.13, 329.63, 349.23, 369.99, 392.00,
    415.30, 440.00, 466.16, 493.88, 523.25, 554.37, 587.33, 622.25,
    659.25, 698.46, 739.99, 783.99, 830.61, 880.00, 932.33, 987.77,
    1046.50
};
const int NUM_NOTES = sizeof(notes) / sizeof(notes[0]);


static uint8_t sineTable[SINE_TABLE_SIZE];
static volatile float currentFreq = 440.0;
static volatile float currentVolume = 0.5;
static volatile bool audioActive = true;
static dac_oneshot_handle_t dac_handle;


static void* display_handle = NULL;


static TaskHandle_t sensorTaskHandle;
static TaskHandle_t audioTaskHandle;
static TaskHandle_t displayTaskHandle;


static ultrasonic_sensor_t pitch_sensor = {
    .trigger_pin = PITCH_TRIG_PIN,
    .echo_pin = PITCH_ECHO_PIN
};

static ultrasonic_sensor_t volume_sensor = {
    .trigger_pin = VOLUME_TRIG_PIN,
    .echo_pin = VOLUME_ECHO_PIN
};
```

```
static void generateSineTable(void);
static float mapRange(float value, float in_min, float in_max, float out_min,
float out_max);
static int findNearestNote(float mappedValue);
static void sensorTask(void *parameter);
static void audioTask(void *parameter);
static void displayTask(void *parameter);
static esp_err_t i2c_master_init(void);
static const char* getNoteName(int noteIndex);

void app_main(void) {
    ESP_LOGI(TAG, "Starting Ultrasonic Theremin with SH1106 Display");

    // Initialize I2C for OLED
    ESP_ERROR_CHECK(i2c_master_init());
    ESP_LOGI(TAG, "I2C initialized");

    // Initialize display
    display_handle = display_init();
    if (display_handle == NULL) {
        ESP_LOGE(TAG, "Failed to initialize display!");
    } else {
        ESP_LOGI(TAG, "Display initialized successfully");



        display_clear(display_handle);
        display_draw_text(display_handle, 25, 20, "THEREMIN", 2);
        display_draw_text(display_handle, 45, 40, "v1.0", 1);
        display_update(display_handle);
        vTaskDelay(pdMS_TO_TICKS(2000));
    }



    esp_err_t res;
    res = ultrasonic_init(&pitch_sensor);
    if (res != ESP_OK) {
```

```
        ESP_LOGE(TAG, "Error initializing pitch sensor: %d", res);
    }

    res = ultrasonic_init(&volume_sensor);
    if (res != ESP_OK) {
        ESP_LOGE(TAG, "Error initializing volume sensor: %d", res);
    }



    generateSineTable();



    dac_oneshot_config_t dac_cfg = {
        .chan_id = DAC_CHAN_0,
    };
    ESP_ERROR_CHECK(dac_oneshot_new_channel(&dac_cfg, &dac_handle));
    ESP_ERROR_CHECK(dac_oneshot_output_voltage(dac_handle, 128)); // Start
at center



    xTaskCreatePinnedToCore(sensorTask,   "SensorTask",   4096,   NULL,   2,
&sensorTaskHandle, 0);
    xTaskCreatePinnedToCore(audioTask,   "AudioTask",   4096,   NULL,   3,
&audioTaskHandle, 1);
    xTaskCreatePinnedToCore(displayTask,  "DisplayTask",  4096,  NULL,  1,
&displayTaskHandle, 0);


    ESP_LOGI(TAG, "Ultrasonic Theremin Started!");


    while(1) {
        vTaskDelay(pdMS_TO_TICKS(1000));
    }
}


static void generateSineTable(void) {
    for (int i = 0; i < SINE_TABLE_SIZE; i++) {
        float angle = (2.0 * M_PI * i) / SINE_TABLE_SIZE;
        sineTable[i] = (uint8_t)((sin(angle) + 1.0) * 127.5);
```

```
    }
}

static float mapRange(float value, float in_min, float in_max, float out_min,
float out_max) {
    if (value < in_min) value = in_min;
    if (value > in_max) value = in_max;
    return (value - in_min) * (out_max - out_min) / (in_max - in_min) +
out_min;
}

static int findNearestNote(float mappedValue) {
    int noteIndex = (int)(mappedValue * (NUM_NOTES - 1));
    if (noteIndex < 0) noteIndex = 0;
    if (noteIndex >= NUM_NOTES) noteIndex = NUM_NOTES - 1;
    return noteIndex;
}

static void sensorTask(void *parameter) {
    const TickType_t taskDelay = pdMS_TO_TICKS(50);

    vTaskDelay(pdMS_TO_TICKS(1000));
    ESP_LOGI(TAG, "Starting Sensor Task");

    uint32_t pitch_distance_cm = 0;
    uint32_t volume_distance_cm = 0;
    esp_err_t pitch_res;
    esp_err_t volume_res;



    const int filterSize = 5;
    float pitch_buffer[filterSize];
    float volume_buffer[filterSize];
    int filter_index = 0;



    for (int i = 0; i < filterSize; i++) {
```

```
        pitch_buffer[i] = 20.0;
        volume_buffer[i] = 20.0;
    }

    while (1) {

        pitch_res = ultrasonic_measure_cm(&pitch_sensor, MAX_DISTANCE,
&pitch_distance_cm);

        vTaskDelay(pdMS_TO_TICKS(10));

        volume_res = ultrasonic_measure_cm(&volume_sensor, MAX_DISTANCE,
&volume_distance_cm);

        if (pitch_res == ESP_OK) {
            pitch_buffer[filter_index] = (float)pitch_distance_cm;
        }

        if (volume_res == ESP_OK) {
            volume_buffer[filter_index] = (float)volume_distance_cm;
        }

        filter_index = (filter_index + 1) % filterSize;

        float avg_pitch = 0, avg_volume = 0;
        for (int i = 0; i < filterSize; i++) {
            avg_pitch += pitch_buffer[i];
            avg_volume += volume_buffer[i];
        }
        avg_pitch /= filterSize;
        avg_volume /= filterSize;
```

```
        if (pitch_res == ESP_OK && pitch_distance_cm <= MAX_DISTANCE) {
            float pitch_mapped = mapRange(avg_pitch, MIN_DISTANCE,
MAX_DISTANCE, 1.0, 0.0);
            int noteIndex = findNearestNote(pitch_mapped);
            currentFreq = notes[noteIndex];
            snprintf(currentNoteName, sizeof currentNoteName, "%s",
getNoteName(noteIndex));
            audioActive = true;
        }
        else {

            audioActive = false;
            strcpy(currentNoteName, "---");
        }

        if (volume_res == ESP_OK) {
            float linear_vol = mapRange(avg_volume, MIN_DISTANCE,
MAX_DISTANCE, 1.0, 0.0);
            currentVolume = linear_vol * linear_vol;
        }

        vTaskDelay(taskDelay);
    }
}

static void audioTask(void *parameter) {
    float phase = 0;
    const int samplePeriodUs = 45;

    ESP_LOGI(TAG, "Starting Audio Task");

    while (1) {
        if (audioActive) {

            int tableIndex = (int)phase % SINE_TABLE_SIZE;
            uint8_t baseSample = sineTable[tableIndex];
```

```
            float scaledSample = 128.0 + (baseSample - 128.0) *
currentVolume;
            uint8_t outputSample = (uint8_t)scaledSample;

            dac_oneshot_output_voltage(dac_handle, outputSample);


            float phaseInc = (currentFreq * SINE_TABLE_SIZE) / SAMPLE_RATE;
            phase += phaseInc;
            if (phase >= SINE_TABLE_SIZE) {
                phase -= SINE_TABLE_SIZE;
            }
        } else {
            dac_oneshot_output_voltage(dac_handle, 128);
            phase = 0;
        }

        esp_rom_delay_us(samplePeriodUs);


        static int yieldCounter = 0;
        if (++yieldCounter >= 500) {
            vTaskDelay(1);
            yieldCounter = 0;
        }
    }
}

static esp_err_t i2c_master_init(void) {
    i2c_config_t conf = {
        .mode = I2C_MODE_MASTER,
        .sda_io_num = I2C_MASTER_SDA_IO,
        .sda_pullup_en = GPIO_PULLUP_ENABLE,
        .scl_io_num = I2C_MASTER_SCL_IO,
        .scl_pullup_en = GPIO_PULLUP_ENABLE,
        .master.clk_speed = I2C_MASTER_FREQ_HZ,
```

```
    };
    esp_err_t err = i2c_param_config(I2C_MASTER_NUM, &conf);
    if (err != ESP_OK) {
        return err;
    }
    return i2c_driver_install(I2C_MASTER_NUM, conf.mode, 0, 0, 0);
}

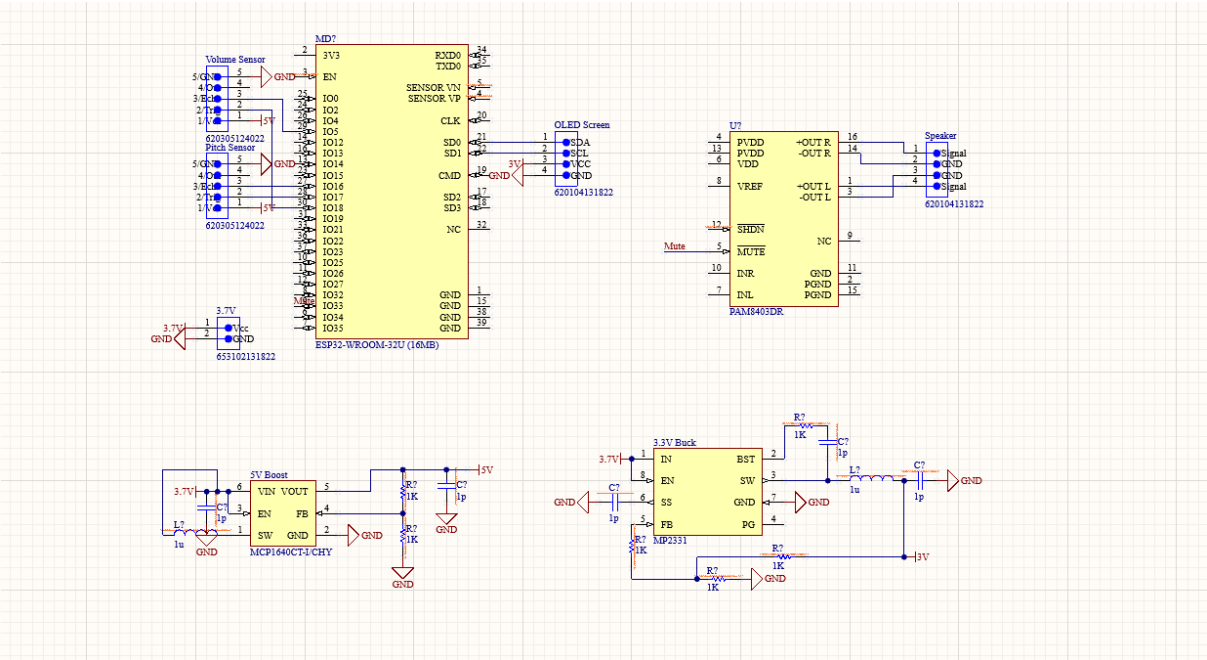
static const char* getNoteName(int noteIndex) {
    const char* noteNames[] = {
        "C4", "C#4", "D4", "D#4", "E4", "F4", "F#4", "G4", "G#4", "A4", "A#4",
"B4",
        "C5", "C#5", "D5", "D#5", "E5", "F5", "F#5", "G5", "G#5", "A5", "A#5",
"B5",
        "C6"
    };

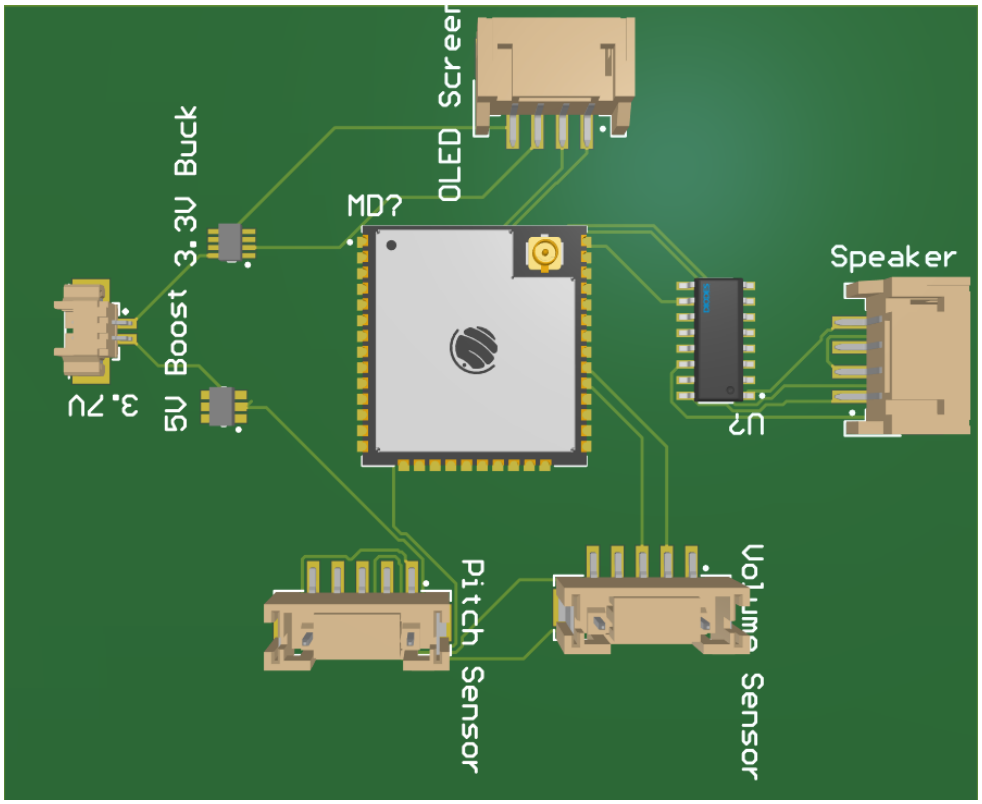
    if (noteIndex >= 0 && noteIndex < NUM_NOTES) {
        return noteNames[noteIndex];
    }
    return "---";
}

static void displayTask(void *arg)
{
    uint16_t phase = 0;
    const TickType_t frameDelay = pdMS_TO_TICKS(50);

    while (1) {
        float cycles = currentFreq / 440.0f;
        display_draw_frame(phase,       currentNoteName,       currentFreq,
currentVolume);
        phase += (uint16_t)(cycles * 4);
        phase &= 0xFFFF;
        vTaskDelay(frameDelay);
    }
}
```

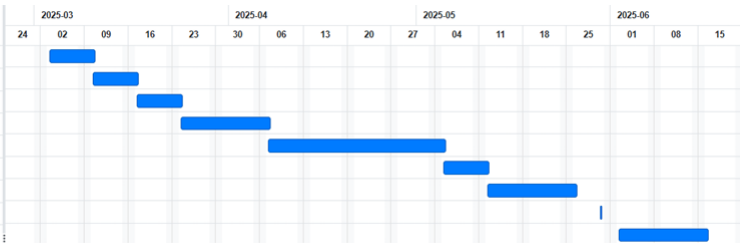
Appendix B: Incomplete Hardware Schematics and PCB Design



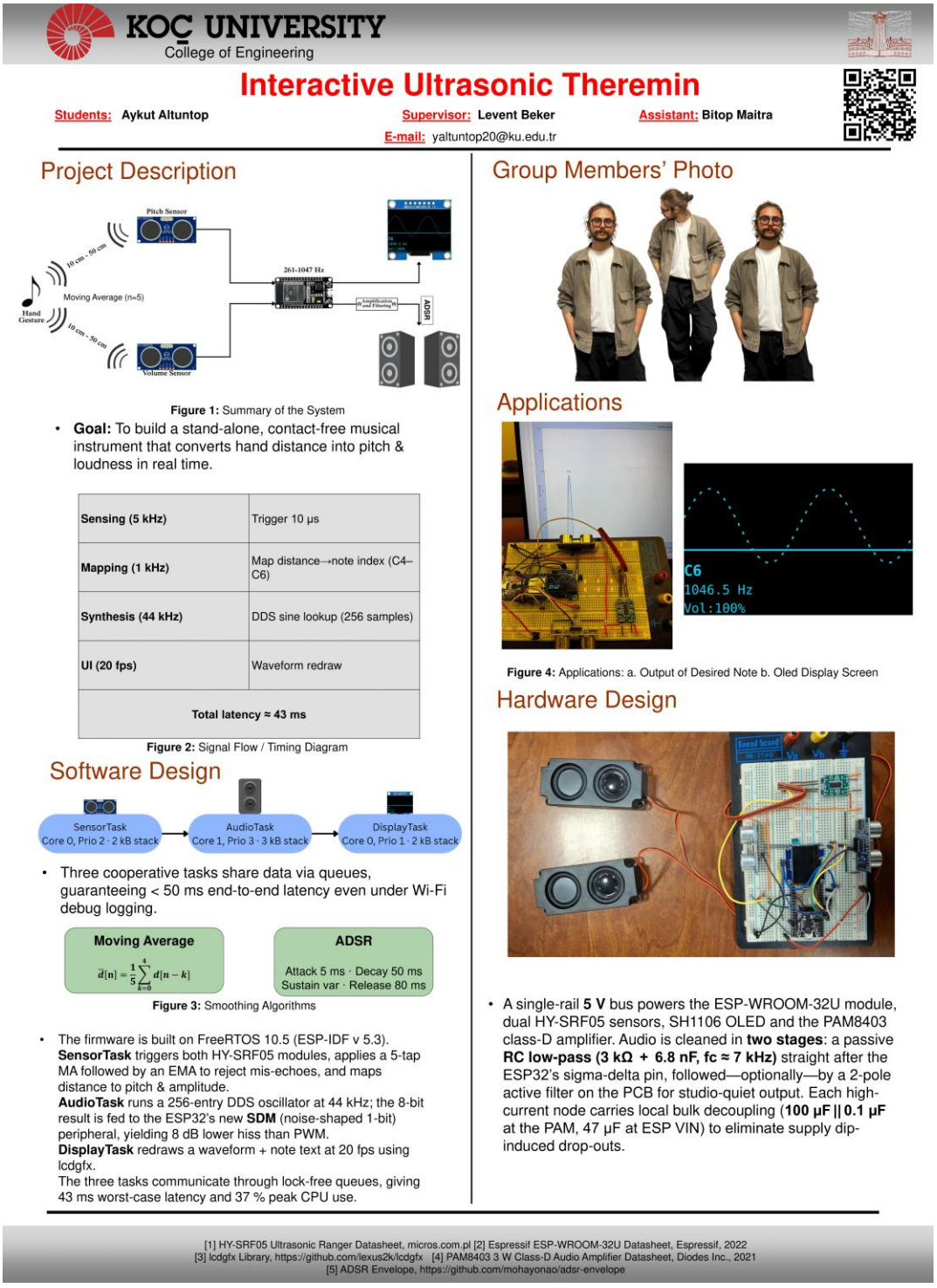


Appendix C: Project Timeline and Development Photos

ID	Task Name	Start	End	Duration
1	Literature Review	2025-03-03	2025-03-10	6 days
2	Initial design of the system	2025-03-10	2025-03-17	6 days
3	Component Acquisition and Algorithm Creation	2025-03-17	2025-03-24	6 days
4	Testing out the components and sound gener…	2025-03-24	2025-04-07	11 days
5	Initial Prototype	2025-04-07	2025-05-05	21 days
6	System adjustments based on tests and feedb…	2025-05-05	2025-05-12	6 days
7	Transition from prototype to designed device	2025-05-12	2025-05-26	11 days
8	Senior Design Day	2025-05-30	2025-05-30	1 day
9	Project Report	2025-06-02	2025-06-16	11 days



Appendix D: Project Poster



Appendix E: Bill of Materials (BOM)

Item	Qty	Unit Price (TL)	Subtotal (TL)	Basic Purpose
ESP32 Dev Board (WiFi+BT, 30 Pin)	1	308.28	308.28	To read sensors, generate audio signals
HY-SRF05 Ultrasonic Sensor (5 Pin)	4	88.02	352.06	To measure distance for pitch control
1/2 W 1 kΩ Resistors (10 pcs)	2	4.15	8.29	To set or filter circuit signals
1/4 W 4.7 kΩ Resistors (10 pcs)	2	1.84	3.69	To use in filter or biasing
1/2 W 10 kΩ Resistors (10 pcs)	2	4.15	8.29	To use in filter or biasing
10 nF Ceramic Capacitor (10 pcs)	2	4.15	8.29	To remove HF noise from signals
100 nF Ceramic Capacitor (10 pcs)	2	8.29	16.59	To remove HF noise from signals
1 μF 50 V Capacitor (per piece)	10	1.38	13.82	For bulk filtering or coupling tasks
8 Ω 5 W Speaker	1	198.61	198.61	To convert electrical waves to sound
40-Pin M–M Jumper (30 cm)	1	43.78	43.78	To interconnect modules on breadboard
40-Pin M–M Jumper (10 cm)	1	52.07	52.07	For additional wiring needs
Large Breadboard	2	52.53	105.07	For temporary prototyping of all components
Marxlow ZD-156F ESD Tweezers	1	92.16	92.16	To safely handle small electronic parts
9 V Battery Clip	1	4.61	4.61	To connect 9 V battery power source
2 A Mini Adjustable Regulator Board	2	31.34	62.67	To maintain stable 3.3-5 V supply level
PAM8403 Class D Amplifier Module	1	15.21	15.21	To amplify MCU audio for speaker
		Grand Total:	**1293.51**	

Table 1: Bill of Materials (BOM) with short purposes.